**To** : M/s HSA MEP **From:** ENGR. METRI

**Attn** : MR. Ahmad Mohammed **Mobile:** +974 3320 4693

**E-mail:** procurement2@hsa-mep.com

**Date** : 1 October, 2026 **Pages** : 03

**Sub** : Design, Supply & Installation of Walk In Chiller room **Ref. #:** 9810-1-rev3

## ENVIRONMENTAL LABORATORY

Dear Sir,

With reference to your inquiry, we are pleased to submit our proposal for your above requirements to wit:

**SCOPE OF WORK** : Design, Supply, installation, Testing & Commissioning of 1 no Walk in Chiller Room to maintain temperature(+4°C) at all times.

### 1nos Walk-in Chiller Room (+4°C )

**EXTERNAL Dimensions** : **3.7 mx 2.2m x 4m Height APPROX. (As per attached location)**
(PUR Wall & PUR Ceiling PANEL THICKNESS **100mm**)

Doors : 1no Single Leaf Hinged Door 0.9m x 1.9m with heater

Floor Insulation : 100mm PUR with checkered aluminum ontop

**Refrigeration Units**: 1no Condensing Unit DORIN ITALY or COPELAND + 1no Matching Evaporator ICEMAKE OR THERMOWAY

**Refrigeration Unit Capacity** : **AU2-H2001CC– Cooling Capacity= 30320 W (see attached)**

**Sandwich Panel INSULATION** : **PUR (POLYURTHANE) (Made in Qatar)**

**THICKNESS** : **100 mm**

**CLADDING** : **0.5 mm pre-painted GI sheets.**

**CONSTRUCTION** : Panels are joined together by means of tongue and groove and locked together on each side of the panel to ensure air tight joints .

**DOOR ACCESSORIES** : Door is complete with fittings and accessories and have the same finish and thickness as that of the walls and ceiling, complete with rubber gaskets, safety release handle.

| | | |
|---|---|---|
| **Lighting** | : | LED lights Included to provide the appropriate light levels for each cold store. |
| **Ref. Accessories** | : | All accessories such as expansion valve, solenoid, sight glass, pressure relief valves …etc are included |
| **CONTROL PANEL** | : | Specially designed to control & secure the system. 1 no. control panel for each cold room will be provided with necessary contactor, overload relay, signal lights…etc |
| **PIPES** | : | copper pipes are made by Muller Brass or Equivalent |
| **M/ ACCESSORIES** | : | Will be provided including rivets, internal and external flashing, silicon and hygienic plastic corners. |
| **Excluding** | : | **Electrical Power to be arranged by client at site & Any Civil Works.** |

## TOTAL PRICE FOR ABOVE + Control Panel = 137,000 Q.R.

| | | |
|---|---|---|
| **PAYMENT TERMS** | : | 60% advance<br>20% upon delivery of cooling units<br>20% upon Completion |
| **DELIVERY** | : | 1 months from the date of your firm order along with the receipt of down payment provided the site is ready for installation. |
| **WARRANTY** | : | 1 year on all materials supplied and installed by MES + free preventive maintenance for the 1st year. Servicing other than MES authorized technician voids warranty. Sandwich Panels 5 years warranty. |

**NOTE :All civil works and main power EXCLUDED. Our work is limited to the above quoted.** *Delay of delivery due to acts of 3rd party (non-availability of supply); site readiness; accidents will not be accounted to MES. Warranty does not cover force majeure, loss due to fire or damage due to natural calamities. MES will not be held liable for any damage on products or items inside the cold room.*

| | | |
|---|---|---|
| **VALIDITY** | : | 30 days. |

We trust the above quotation merits your attention and we look forward to serving your requirement soon. Should you need any further clarification, please do not hesitate to contact us.

Thanks & regards,

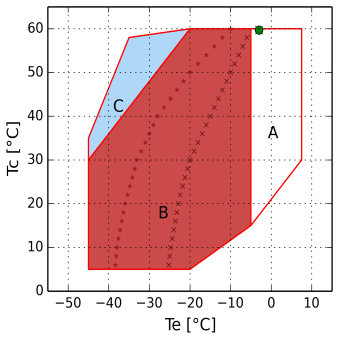
**Metri Engineering Services W.L.L**



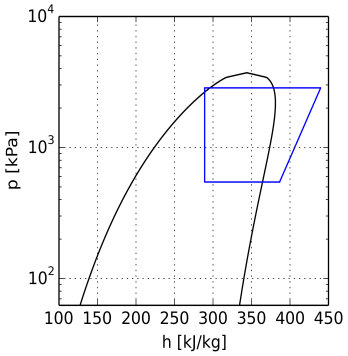
# Calculation - AU

## Input

| | |
|---|---|
| Model | AU2-H2001CC |
| Refrigerant | R404A |
| Evaporating temperature | -3.0 °C |
| Ambient temperature | 46.0 °C |
| Reference temperature | Dew temperature |
| Suction temperature | 20.0 °C |
| Evaporator outlet temperature | 20.0 °C |
| Voltage / phases / frequency | 380-420 V / 3 / 50 Hz |




A = "CC" models only
B = Standard application
C = Head cooling fan or max superheating 20K
× = Min Te 50%
⋆ = Min Te 50% Head cooling fan or max superheating 20K
• = Dew temperature



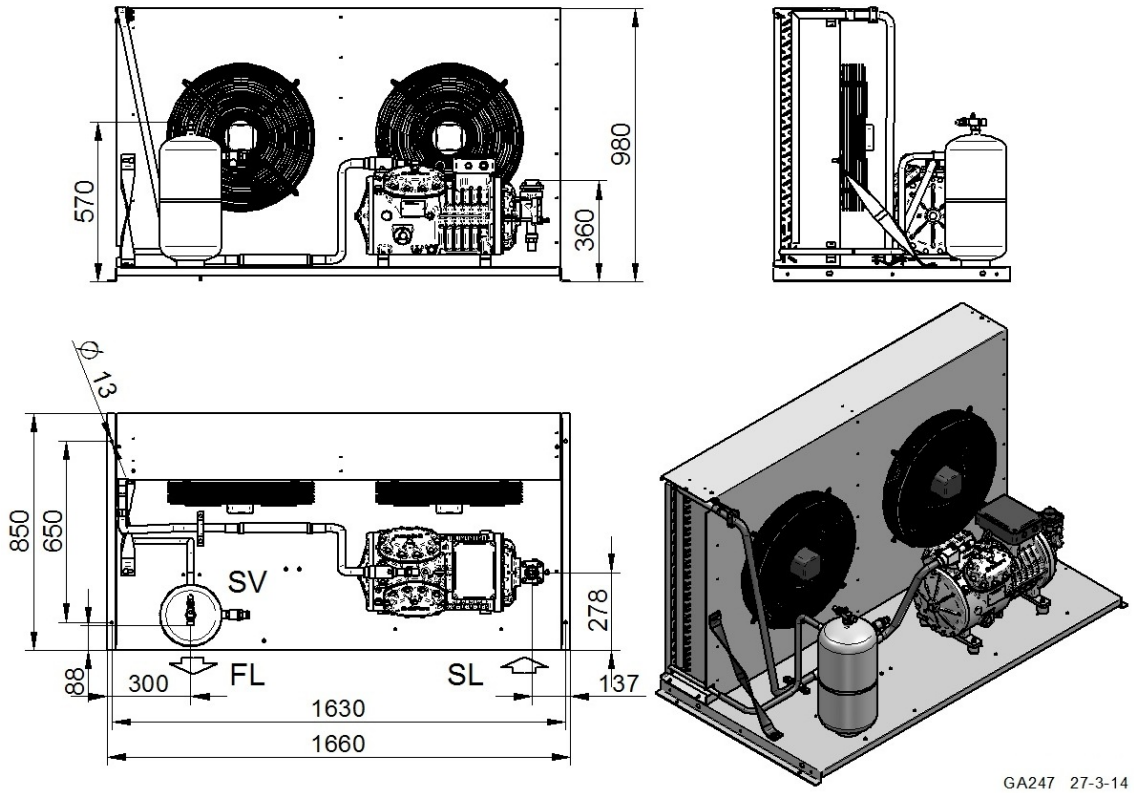

## Performance calculation

| | Standard conditions | At evaporator | At compressor |
|---|---|---|---|
| Cooling capacity | 28170 W | 30320 W | 30320 W |
| Absorbed power including fans | 18.45 kW | 18.45 kW | 18.45 kW |
| Condenser capacity | 46.61 kW | 47.40 kW | 48.76 kW |
| COP | 1.53 | 1.64 | 1.64 |
| Mass flow | 1138.8 kg/h | 1138.8 kg/h | 1138.8 kg/h |
| Condensing temperature | 59.7 °C | 59.7 °C | 59.7 °C |
| Liquid subcooling | 0.0 K | 3.0 K | 3.0 K |
| Absorbed current | 30.0 A | 30.0 A | 30.0 A |
| Max operating current | 38.0 A | 38.0 A | 38.0 A |
| Locked rotor current | 177.0 A | 177.0 A | 177.0 A |

Technical data

| | | |
|---|---|---|
| Compressor model | H2001CC | |
| Displacement @ 50 Hz | 56.87 | [m³/h] |
| Displacement @ 60 Hz | 68.25 | [m³/h] |
| Motorfans: Air Flow (tot.) @50Hz | 14500.0 | [m³/h] |
| Motorfans: Air Flow (tot.) @60Hz | 16320.0 | [m³/h] |
| Motorfans: Power @50Hz | 2×680 | n° × [W] |
| Motorfans: Power @60Hz | 2×880 | n° × [W] |
| Motorfans: Current consumption (tot.) @50Hz | 6.0 | [A] |
| Motorfans: Current consumption (tot.) @60Hz | 7.76 | [A] |
| Motorfans: Condenser capacity @50Hz | 2×12 | n° × µF |
| Motorfans: Condenser capacity @60Hz | 2×12 | n° × µF |
| Motorfans | 2×500 | n° × Ø |
| Receiver volume | 15.0 | [L] |
| Connections: Suction | 42 s. | [mm] |
| Connections: Liquid | 7/8" | |
| Net weight | 257.0 | [kg] |
| Sound pressure (free field) 1,8 m @50Hz | 64.1 | [dBA] |
| Sound pressure (free field) 1,8 m @60Hz | 65.3 | [dBA] |
| Sound pressure (free field) 5 m @50Hz | 55.2 | [dBA] |
| Sound pressure (free field) 5 m @60Hz | 56.4 | [dBA] |
| Sound power @50Hz | 80.2 | [dBA] |
| Sound power @60Hz | 81.4 | [dBA] |

Overall dimensions [mm]




FL - Refrigerant outlet

SV - Safety valve

SL - Suction service valve